WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

Numeral Systems

Lecture 02: additive and positional systems; binary, octal and hexadecimal

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01

Ways of counting

Numbers and numerals

Counting is an unavoidable part of human life, and people have invented many different ways of writing counts down.

> **Definition (Numeral system)**
>
> A *numeral system* is a way of writing numbers with symbols. A *numeral* is one number written in that way.

One number, four numerals:

tally	Roman	decimal	binary
||||||	VI	6	110_2

There are two big families: additive and positional systems.

Additive systems

In an additive system, each symbol has a fixed value, and the number is the sum of the symbols. Where a symbol stands does not change its magnitude.

Roman numerals

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

- VI = 5 + 1 = 6
- IV = −1 + 5 = 4: a smaller symbol before a larger one is subtracted, but its magnitude is still 1
- MMXXVI = 1000 + 1000 + 10 + 10 + 5 + 1 = 2026

Tally marks are additive too: every mark is worth 1, wherever it is.

The trouble with additive systems

- There is no symbol for zero.
- Larger numbers need new symbols, or very long numerals:
  3888 = MMMDCCCLXXXVIII takes 15 symbols.
- Written arithmetic is awkward. Try multiplying XLVII by XXXVIII on paper.

> **Remark**
>
> Positional systems are far more useful in almost every area of mathematics and in digital logic design. From now on we work only with them.

Positional systems

In a positional system, the place of a digit decides its magnitude.

$$\begin{aligned} 123 &= 100 + 20 + 3 \\ 321 &= 300 + 20 + 1 \end{aligned}$$

The same three digits, three different numbers.

- Zero holds a place: 105 and 15 differ only by the 0.
- Our decimal digits are positional with ten digits. Other choices are just as possible.

> **Example (Base 60, still in use)**
>
> The Babylonians counted in base 60, and we still do for time: 1 h 15 min 30 s is $1 \cdot 60^2 + 15 \cdot 60 + 30 = 4530$ seconds.

02

The radix

Radix and digits

Definition (Radix)

The *radix*, or *base*, b of a positional system is the number of elementary symbols, the *digits*, that all its numerals are made of. The digits stand for $0, 1, \ldots, b-1$.

Each digit is multiplied by the power of b for its position, counted from 0 on the right, and the results are added:

$$\overline{d_n d_{n-1} \ldots d_0}_{\,b} = d_n \cdot b^n + d_{n-1} \cdot b^{n-1} + \cdots + d_0 \cdot b^0.$$

Example (Decimal, $b = 10$)

$2026 = 2 \cdot 10^3 + 0 \cdot 10^2 + 2 \cdot 10^1 + 6 \cdot 10^0$

Binary, octal and hexadecimal

System	Radix	Digits	Example
binary	2	0, 1	$101101_2 = 45$
octal	8	0, 1, …, 7	$55_8 = 45$
decimal	10	0, 1, …, 9	$45_{10} = 45$
hexadecimal	16	0, 1, …, 9, A, B, C, D, E, F	$2\mathrm{D}_{16} = 45$

- Hexadecimal needs sixteen digits, for the values 0 to 15. After 9 come the letters: A = 10, B = 11, …, F = 15.
- The value formula is the same for every radix: $\overline{b_n \dots b_0}_{\,2} = b_n \cdot 2^n + \cdots + b_0$, and so on.

Counting in four bases

dec	bin	oct	hex
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7

dec	bin	oct	hex
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

In every base, counting works like an odometer: when a digit passes its largest value, it goes back to 0 and the next digit goes up by one ($1111_2 + 1 = 10000_2 = 16$).

Why binary?

A switch is on or off: two states, so two digits (Lecture 01).

- One binary digit is a bit. Eight bits are a byte; four bits are a *nibble*, exactly one hexadecimal digit.
- n bits have 2^n states, so they can stand for the numbers 0 to $2^n - 1$.

n	0	1	2	3	4	5	6	7	8	9	10
2^n	1	2	4	8	16	32	64	128	256	512	1024

How many bits do you need for the numbers 0 to 999?

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How many bits do you need for the numbers 0 to 999? $2^9 = 512$ is too few and $2^{10} = 1024$ is enough: 10 bits.

Writing the base

The digits alone do not say which base is meant: is 10 ten, two, eight or sixteen?

Where	Binary	Octal	Decimal	Hexadecimal
these slides	1011_2	13_8	11	B_{16}
many programming languages	`0b1011`	`0o13`	`11`	`0xB`
Verilog (4 bits)	`4'b1011`	`4'o13`	`4'd11`	`4'hB`

The book writes $\overline{d_n \ldots d_0}_{\,b}$, with a bar over the digits. Numbers without a base are decimal.

03

Converting between bases

To decimal: expand

Multiply each digit by its power of the base and add.

$$101101_2 = 1 \cdot 32 + 0 \cdot 16 + 1 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 45$$
$$57_8 = 5 \cdot 8 + 7 = 47$$
$$2\mathrm{F}_{16} = 2 \cdot 16 + 15 = 47$$

> **A shortcut for binary**
>
> Every binary digit is 0 or 1, so just add up the powers of 2 where the bits are 1: $101101_2 = 32 + 8 + 4 + 1 = 45$.

To decimal: Horner's rule

Read the digits from the left. Keep one number v, starting at 0, and for each digit d: $v \leftarrow v \cdot b + d$.

Digit	v	
1	1	$= 0 \cdot 2 + 1$
0	2	$= 1 \cdot 2 + 0$
1	5	$= 2 \cdot 2 + 1$
1	11	$= 5 \cdot 2 + 1$
0	22	$= 11 \cdot 2 + 0$
1	45	$= 22 \cdot 2 + 1$

- Converting 101101_2: no powers to compute, one multiplication and one addition per digit.
- A computation in the sense of Lecture 01: the state is v, the input is the next digit.

It works because the powers can be factored out:

$$101101_2 = ((((1 \cdot 2 + 0) \cdot 2 + 1) \cdot 2 + 1) \cdot 2 + 0) \cdot 2 + 1$$

From decimal: repeated division

Divide by the base and write down the remainder; repeat with the quotient until it is 0. The remainders, read from the last to the first, are the digits.

N	$N \div 2$	remainder
45	22	1
22	11	0
11	5	1
5	2	1
2	1	0
1	0	1

- Read upwards: $45 = 101101_2$.
- Why it works: $N = q \cdot b + d_0$, so the remainder is the last digit d_0, and the quotient q has the remaining digits $d_n \ldots d_1$.
- The same method works for any base.

Example: 2026 in hexadecimal

N	$N \div 16$	remainder	
2026	126	10	A
126	7	14	E
7	0	7	7

Read upwards: $2026 = 7\text{EA}_{16}$.

Check by expanding:

$$7 \cdot 256 + 14 \cdot 16 + 10 = 1792 + 224 + 10 = 2026.$$

Common mistake

Reading the remainders from the top gives AE7_{16}, which is 2791. The first remainder is the *last* digit.

Binary to octal and hexadecimal: group the bits

Since $8 = 2^3$ and $16 = 2^4$, every octal digit is exactly 3 bits and every hexadecimal digit exactly 4 bits. Group the bits from the right.

Take 1011010111_2:

groups of 4		0010	1101	0111	
hexadecimal		2	D	7	$= 2D7_{16}$
groups of 3	001	011	010	111	
octal	1	3	2	7	$= 1327_8$

All three are 727. Pad the leftmost group with zeros if it is short.

And back again

- Hexadecimal to binary: write each digit as 4 bits.
  $C3_{16} = 1100\ 0011 = 11000011_2 = 195$
- Octal to binary: write each digit as 3 bits.
  $57_8 = 101\ 111 = 101111_2 = 47$
- Octal to hexadecimal, or back: go through binary.
  $1327_8 = 1011010111_2 = 2D7_{16}$

> **Remark**
>
> This is why hexadecimal is everywhere in computing: it is a short way of writing bits. One byte is always two hexadecimal digits, from 00_{16} to FF_{16}.

Try it

1. Write 100 in binary, octal and hexadecimal.
2. What is FF_{16} in binary and in decimal?
3. Which is larger: 1000000_2 or 77_8?

Try it

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   Answer: $100 = 64 + 32 + 4 = 1100100_2$; $\texttt{1 100 100} = 144_8$; $\texttt{0110 0100} = 64_{16}$.
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Try it

1. Write 100 in binary, octal and hexadecimal.
   `Answer:` $100 = 64 + 32 + 4 = 1100100_2$; `1 100 100` $= 144_8$; `0110 0100` $= 64_{16}$.
2. What is FF_{16} in binary and in decimal?
   `Answer:` $11111111_2 = 2^8 - 1 = 255$, the largest number in one byte.
3. Which is larger: 1000000_2 or 77_8?
   `Answer:` $1000000_2 = 64$ and $77_8 = 63$, so 1000000_2, by one.

04

Numbers in the DX8

Numbers in the DX8

The computer we build in this module works with bits everywhere:

- it is an 8-bit computer: one byte holds a number from 0 to 255;
- its 8 registers are selected by 3-bit addresses, 000_2 to 111_2, because $2^3 = 8$;
- its RAM has $2^8 = 256$ bytes, so a memory address is one byte, 00_{16} to FF_{16};
- an instruction is 16 bits: four hexadecimal digits.

> **Coming back**
>
> So far every number is 0 or positive. Negative numbers in binary, and how to add them, come with two's complement `(Unit 16)`.

05

Summary

Summary

- In an additive system each symbol has a fixed value; in a positional system the place of a digit decides its magnitude.
- With radix b: digits $0, \ldots, b-1$ and $\overline{d_n \ldots d_0}_b = \sum_i d_i \cdot b^i$.
- To decimal: expand, or use Horner's rule $v \leftarrow v \cdot b + d$. From decimal: divide repeatedly, read the remainders from the last.
- Binary to octal or hexadecimal: group the bits by 3 or by 4, from the right.
- n bits give 2^n numbers, from 0 to $2^n - 1$.

Before the next lecture

- Read Section 2.1 of the book, *Numeral Systems*.
- Practise: write your year of birth in binary, octal and hexadecimal, and check each answer by converting it back.
- Count from 0 to 31 in binary. Which bit changes at every step, and which only once?

> **Next lecture** (Unit 3)
>
> Boolean algebra: an algebra with only the values 0 and 1, and its laws and theorems.